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 Mata Kuliah : Struktura Data dan Algorithma
 Tugas : 2 (Hash Function & Hash Table)

① $H(k) = k \bmod 11$

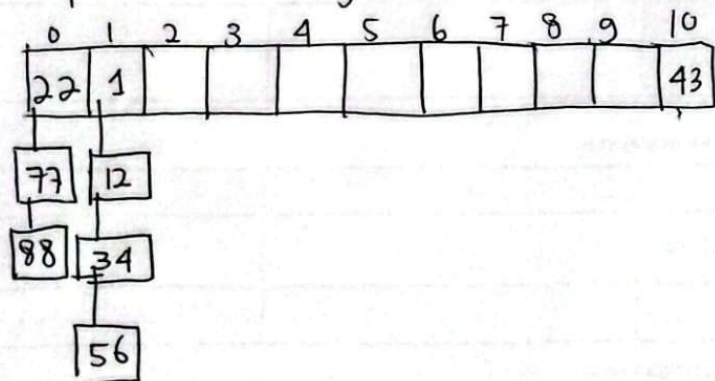
keys = [43, 22, 1, 12, 34, 56, 77, 88]

a. Linear Probing (mencari slot kosong berikutnya)

keys	module 11	posisi
43	$\frac{43}{11} = 3$ sisa 10 $\rightarrow (10)$	
22	$\frac{22}{11} = 2$ sisa 0 $\rightarrow (0)$	
1	$\frac{1}{11} = 0$ sisa 1 $\rightarrow (1)$	
12	$\frac{12}{11} = 1$ sisa 1 $\rightarrow (2)$	
34	$\frac{34}{11} = 3$ sisa 1 $\rightarrow (3)$	
56	$\frac{56}{11} = 5$ sisa 1 $\rightarrow (4)$	
77	$\frac{77}{11} = 7$ sisa 0 $\rightarrow (5)$	
88	$\frac{88}{11} = 8$ sisa 0 $\rightarrow (6)$	

0	1	2	3	4	5	6	7	8	9	10
22	1	12	34	56	77	88				43

b. Separate Chaining



```

2. public class HashTable {
    static class Node {
        int key;
        int value;
        Node next;

        Node(int k, int v) {
            this.key = k;
            this.value = v;
        }
    }

    private Node[] tabel = new Node[10];

    public void insert(int key) {
        int indeks = key % 10;
        int value = key % 10;

        Node newNode = new Node(key, value);
        newNode.next = tabel[indeks];
        tabel[indeks] = newNode;
    }

    public String search(int key) {
        int indeks = key % 10;
        Node temp = tabel[indeks];

        while (temp != null) {
            if (temp.key == key) {
                return "Ditemukan! Value: " + temp.value;
            }
        }
    }
}

```

```

        temp = temp.next;
    }
    return "Tidak ditemukan";
}

public void display() {
    System.out.println("\n=== HASH TABLE ===");
    for (int i = 0; i < 10; i++) {
        System.out.print("Indeks " + i + ": ");
        Node temp = tabel[i];

        if (temp == null) {
            System.out.print("kosong");
        }

        while (temp != null) {
            System.out.print("[ " + temp.key + " -> Val: " + temp.value + " ]");
            if (temp.next != null) {
                System.out.print(" -> ");
            }
            temp = temp.next;
        }
        System.out.println();
    }
    System.out.println("=====\n");
}

public static void main(String[] args) {
    HashTable ht = new HashTable();

    int[] dataKeys = {43, 22, 1, 12, 34, 56, 77, 88};

```

```

for (int k : dataKeys) {
    ht.insert(k);
}
ht.display();
Scanner input = new Scanner(System.in);
while (true) {
    System.out.print("Masukkan key yang dicari (atau ketik -1 untuk keluar): ");
    int keyDicari = input.nextInt();

    if (keyDicari == -1) {
        System.out.println("Program selesai. Terima kasih!");
        break;
    }

    String hasilPencarian = ht.search(keyDicari);
    System.out.println("Hasil untuk key " + keyDicari + ": " + hasilPencarian);
    System.out.println("-----");
}
}
}

```

OUTPUT:

```

=== HASH TABLE ===
Indeks 0: kosong
Indeks 1: [1 -> Val: 1]
Indeks 2: [12 -> Val: 2] -> [22 -> Val: 2]
Indeks 3: [43 -> Val: 3]
Indeks 4: [34 -> Val: 4]
Indeks 5: kosong
Indeks 6: [56 -> Val: 6]
Indeks 7: [77 -> Val: 7]
Indeks 8: [88 -> Val: 8]
Indeks 9: kosong
=====

Masukkan key yang dicari (atau ketik -1 untuk keluar): 22
Hasil untuk key 22: Ditemukan! Value: 2
-----

Masukkan key yang dicari (atau ketik -1 untuk keluar): 13
Hasil untuk key 13: Tidak ditemukan
-----

```

Github: [marisanelisabeth-lgtm](#)

